

substrate that shields developers from the complexities of the underlying cloud platform.

Q. How is this scalability achieved and how does it impact the setup cost?

A. This scalability is partly because running Cloud Foundry requires a substantial setup—typically 30 to 50 VMs, which enables it to optimise compute resources at scale. It can handle a high volume of applications and deploy numerous workloads and instances as needed. It is cloud-agnostic; it works well across various cloud providers, with versions available for Azure, AWS, Google Cloud, SAP, and formerly IBM and SUSE. This flexibility makes it easy to deploy on your chosen cloud provider through its marketplace offerings.

Q. How does this impact the hardware infrastructure costs?

A. It's not something that can simply run on a laptop, which does present challenges, but with that scale of resources, Cloud Foundry is highly optimised. For example, Cloud Foundry applies the principle of 'bin packing'—a feature in Kubernetes designed to optimise compute for container clusters—but it applies this principle to VMs, ensuring efficient use of every VM deployed. It operates on VMs, so you aren't tied to specific hardware, though VMs do come with associated costs.

Q. How does Cloud Foundry handle DevOps automation?

A. Automation in Cloud Foundry relies on two core aspects: capabilities and features that allow tools to be automated, and the ability to seamlessly move processes from one stage to the next. Cloud Foundry was designed with both in mind, making it essential for many DevOps automation needs. It simplifies common workflows—like

building, deploying, scaling, logging, monitoring, and applying security best practices—into a single command.

For instance, if you have your code on your laptop, you can configure a Cloud Foundry (CF) endpoint on a remote cloud provider. By executing `cf push <app-name>`, Cloud Foundry takes the source code, uploads it to the cloud controller, builds it, attaches necessary services, deploys it to production, and then provides a URL to access the app—all in one step.

Additionally, Cloud Foundry offers various services (e.g., databases, message queues, pub/sub tools like Kafka) through the CF Marketplace, which you can attach with a single command. Instead of creating databases, handling credentials, or manually configuring connections, you can simply use `cf bind-service` to link these services to your app.

These single-command operations integrate seamlessly with any automation tool, enabling efficient and streamlined workflows for DevOps teams.

Q. What kind of security measures or checks and balances are in place to ensure that the vulnerabilities are properly taken care of?

A. Cloud Foundry takes a multi-pronged approach to managing security within the foundation. We have a Vulnerability Management Working Group that monitors vulnerabilities across all Cloud Foundry projects. While no project is entirely immune to security issues—Cloud Foundry was also affected by incidents like the Log4j vulnerability—the working group provides ongoing oversight to mitigate and resolve issues promptly.

We also have stem cells, which are the OS-level base that supports workloads built on top of the platform. Keeping stem cells secure, low-risk, and up-to-date offers an additional layer of protection, as they provide a stable,

secure foundation for all applications deployed within Cloud Foundry. Likewise, the use of Buildpacks within Cloud Foundry makes it easier to patch a large number of containers. The community's vigilance and contributions are invaluable in maintaining the integrity and functionality of Cloud Foundry.

Q. How do you see the future of the Cloud Foundry project evolving?

A. The foundation is focused on maintaining the current Cloud Foundry on VMs and its existing ecosystem, while also working to extend Cloud Foundry to newer technologies like Kubernetes. Beyond Cloud Foundry, the foundation supports related projects like Buildpacks and the Open Service Broker API, which are relevant across multiple ecosystems. Our goal is to back these projects, nurturing and sustaining the communities around them.

In addition to Cloud Foundry, we engage in various initiatives under the Linux Foundation umbrella. For instance, we integrate best practices from the Open Source Security Foundation and collaborate with the CNCF, adopting beneficial tools and practices, such as Buildpacks, where they fit. By leveraging these synergies within the Linux Foundation, we aim to grow and evolve alongside the broader open source ecosystem.

Q. How has your experience been at Open Source India (OSI) 2024?

A. I really think we should have brought in a sponsorship for this year's event, but we'll definitely work on something early enough before next year's edition. I really loved being part of this huge exhibition of open source tech; there's a lot more we can do as an open source foundation with the OSI community and take advantage of a lot of the OSI bells and whistles. **END** 🐧